HeartWise — Heart Disease Prediction Using Machine Learning

A Streamlit portfolio application demonstrating dataset exploration, preprocessing pipelines, model comparison, evaluation metrics, and a prediction interface.

Important dataset note

The built-in offline dataset is scikit-learn's breast-cancer dataset, included only so the app runs without downloading a file. It is **not a heart-disease dataset**. For valid heart-disease analysis, upload a documented heart-disease CSV and verify its feature meanings and target coding.

Features

• Built-in demo dataset and user CSV upload

• Dataset preview, missing-value summary, class counts, distributions, correlation heatmap

• Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and KNN

• Accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, ROC curve

• Interactive model prediction demo

• CSV download

• Medical-use warning

Run locally

```
python -m venv .venv
# Windows: .venv\\Scripts\\activate
# macOS/Linux: source .venv/bin/activate
pip install -r requirements.txt
streamlit run app.py
```

Deploy on Streamlit Community Cloud

• Upload app.py, requirements.txt, and this README to a GitHub repository.

• Open Streamlit Community Cloud and create an app from that repository.

• Set the main file path to app.py.

CSV upload format

• Include one binary target column named target, num, heart_disease, disease, output, or label.

• Target values should be 0 and 1 (1 is the positive class).

• Numeric feature columns are used for modelling.

• Do not upload identifiable patient information.

Responsible use

This project is for education and portfolio demonstration only. It is not a medical device and must not be used to diagnose, treat, or rule out disease. A qualified clinician should interpret health concerns.